Sabarishwaran G

sabarish261101@gmail.com — +91 8610605084 — LinkedIn — Trichy, Tamil Nadu

Skills

Languages & Tools: Python, SQL (MySQL, BigQuery), Git, GitHub, Docker, REST APIs, Bash
Machine Learning & AI: Machine Learning, Deep Learning, Transformers, Autoencoders, NLP, Computer Vision, Reinforcement Learning, LLMs, LightGBM, XGBoost, PyTorch, Scikit-learn
Statistical Analysis: SHAP Analysis, Statistical Modeling, Linear Algebra, A/B Testing
Data Processing: Exploratory Data Analysis, Data Wrangling, Feature Engineering, Apache Arrow
Visualization: Matplotlib, Seaborn, Plotly
Cloud & MLOps: AWS (SageMaker, EC2, S3, EMR, Athena), GCP (Vertex AI Pipelines, BigQuery, Cloud Run, GCS, Cloud Scheduler, IAM), Kubeflow Pipelines (KFP), Terraform
Big Data: PySpark, AWS EMR, BigQuery Storage API

Work Experience

Data Scientist — Trellix *July 2023 – Present*

- **ML-Powered Threat Detection for EDR Systems**
  - Replaced high-volume, manually managed YARA rules with a supervised ML model classifying EDR telemetry into **353 rule types**, achieving **94.5% weighted F1**.
  - Built a feature pipeline transforming raw telemetry into behavioral sentences and **384-dim transformer embeddings**.
  - Improved detection quality via targeted retraining on misclassified samples, boosting **57 rules to 90% precision**; integrated an LLM to generate detection insights and MITRE ATT&CK mappings.
  - **Technologies:** Python, LightGBM, Scikit-learn, Transformers, AWS (EC2, SageMaker, S3)
- **End-to-End MLOps Pipeline for Malware Classification (GCP)**
  - Owned a production 7-stage ML pipeline (inventory → labeling → extraction → sampling → HP tuning → training → evaluation) on Vertex AI Pipelines and Cloud Run Jobs, classifying Windows PE files over **2,598+ extracted features**.
  - Resolved conflicting labels across **6 upstream sources** (VirusTotal, ReversingLabs, AVTest, DoD, endpoint feedback, manual analysts) via a tiered combiner with versioned label logic; chronological train/val/test splits prevent temporal leakage.
  - Trained LightGBM on Vertex AI GPU CustomJobs with a 3-tier hyperparameter fallback (Bayesian HP-tuning winner → config defaults → library defaults); tuned 8 params with parallel Bayesian trials.
  - Scaled to **120M+ row** training sets via Arrow streaming over the BigQuery Storage Read API, dtype compaction (f64→f32, i64→i32), and Parquet split caching on GCS.
  - Automated infrastructure with Terraform (BQ datasets, GCS buckets, IAM, daily Cloud Scheduler retrains) and decoupled config from images via GCS-hydrated YAML.
  - **Technologies:** Python, LightGBM (GPU/OpenCL), GCP (Vertex AI, BigQuery, Cloud Run, GCS, Cloud Scheduler, IAM), Kubeflow Pipelines, Terraform, Docker, Apache Arrow
- **Alert Prioritization & Threat Detection**
  - Designed and implemented an AutoEncoder-based alert prioritization system after extensive research into industry best practices.
  - Reduced false positives from **70% to 30%** and cut alert fatigue by **94.5%**, significantly improving SOC efficiency.
  - **Technologies:** Python, PyTorch, Scikit-learn, AutoEncoders, PySpark, Git, AWS (EC2, SageMaker, S3, EMR, Athena)

Data Science Intern — Trellix *Mar 2023 – Jun 2023*

- **URL Categorization using Machine Learning**
  - Developed an automated URL categorization pipeline using NLP-based content analysis.
  - Generated multilingual embeddings with **XLM transformers** and trained an **XGBoost** model achieving **76% accuracy** across 60 categories.
  - Applied **SHAP** to interpret model predictions and validate feature importance.

Education

B.Tech in Computer Science Engineering (Artificial Intelligence) *Sept 2019 – May 2023*
Amrita Vishwa Vidyapeetham, Coimbatore
CGPA: 8.02 / 10